Geonu Lee (이건우)

AI Research Engineer

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Work Experiences

Jul. 2024 - Present

Full-time – SNUAILAB

AI Research Engineer

- **Led the end-to-end development and deployment** of an AI-based smartphone cover glass micro-defect inspection system, achieving a False Negative Rate (FNR) below 2%.
- **Managed technical alignment and crisis resolution** with a manufacturing client, establishing technical trust that successfully secured strategic corporate investment.
- **Trained deep learning models** for a battery pole tracking system and integrated real-time pipelines into a C++ SDK, stabilizing edge deployment for a specialized inspection equipment manufacturer.
- **First-authored** "Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection" (*Neurocomputing*, Under Minor Revision).
- **Submitted a first-author paper on 3D anomaly detection** to *ECCV 2026* (Under Rebuttal).

Oct. 2023 - Apr. 2024

Full-time – ALCHERA

AI Research Engineer, Anomaly Analysis Team

- **Developed and deployed DETR-based object detection models** for FireScout, a real-time vision-based wildfire detection solution, achieving a 20%p performance improvement over the legacy baseline.
- **Reconfigured the evaluation pipeline** from image-level to instance-level metrics, resolving legacy evaluation limitations to establish a precise and robust performance measurement system.

Mar. 2023 - Sep. 2023

Intern – Naver Cloud

AI Research Engineer, Image Vision Team

- **Contributed to the development of a face anti-spoofing service framework** for eKYC services, constructing a robust system capable of detecting spoofing attacks across diverse environmental conditions.
- **First-authored a paper published at WACV 2025** on object anti-spoofing, improving HTER by 7.5% and EER by 6.6% to significantly enhance domain generalization performance.

Jan. 2020 - Feb. 2020

Intern – Electronics and Telecommunications Research Institute (ETRI)

AI Research Engineer

- Developed a human action recognition system that models interactions between humans and surrounding objects in video data.

Research Interests

Deep Learning, Pose Estimation, Multi-Modal, Vision-Language-Model, Vision-Language-Action, Anomaly Detection, Domain Generalization, Face Anti-Spoofing, Human Action Recognition, Human-Object Interaction Detection

Publications

May. 2026 | **Geonu Lee**, YoungJoon Yoo, “[Title Anonymized] A Study on 3D Anomaly Detection,” in *European Conference on Computer Vision (ECCV)* (Under rebuttal)

Feb. 2026 | **Geonu Lee**, Yujeong Oh, Geonhui Jang, Soyoung Lee, Jeonghyo Song, Sungmin Cha, YoungJoon Yoo, “Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection,” in *Neurocomputing* (Minor Revision)

Feb. 2025 | **Geonu Lee**, Yonghyun Jeong, Haneol Jang, YoungJoon Yoo, “Domain-Generalized Object Anti-Spoofing: Bridging Gaps and Patch Selection for Robust Detection across Domains,” in *Winter Conference on Applications of Computer Vision (WACV)*

Sep. 2022 | **Geonu Lee**, Kimin Yun, Jungchan Cho, “Occluded Pedestrian-Attribute Recognition for Video Sensors Using Group Sparsity,” in *Sensors*, vol. 22, no. 17, pp. 6626

Aug. 2022 | **Geonu Lee** and Jungchan Cho, “STDP-Net: Improved Pedestrian Attribute Recognition Using Swin Transformer and Semantic Self-Attention," in *IEEE ACCESS*, vol. 10, no.1, pp. 82656 - 82667

Feb. 2021 | **Geonu Lee**, Kimin Yun, and Jungchan Cho, “Improved Human-Object Interaction Detection through On-the-Fly Stacked Generalization," in *IEEE ACCESS*, vol. 9, no. 1, pp. 34251-34263

Feb. 2020 | Bhishan Bhandari, **Geonu Lee**, and Jungchan Cho, “Body-Part-Aware and Multitask-Aware Single-Image-Based Action Recognition," in *Applied Science*, vol. 10, no. 4, pp. 1531-1548

Research Projects

2022 | **A Study on the Complex Human Attributes for Situation Understanding**
Master’s Research Project – Gachon University
- Conducted research on recognizing interactive human attributes in video sequences under complex contexts.
- Utilized Swin Transformer and Transformer decoder architectures for semantic reasoning.
- Funded by Electronics and Telecommunications Research Institute (ETRI).

2021 | **A Study on the Understanding of Pedestrians**
Master’s Research Project – Gachon University
- Developed a pedestrian attribute recognition method robust to occlusion.
- Applied group sparsity regularization to handle various levels of visual obstruction.
- Funded by Electronics and Telecommunications Research Institute (ETRI).

2020 | **A Study on the Understanding of Human-Object-Interactions**
Undergraduate Research Project – Gachon University
- Designed a novel deep neural architecture based on on-the-fly stacked generalization for HOI detection.
- Focused on modeling dynamic interactions between humans and surrounding objects.
- Funded by Electronics and Telecommunications Research Institute (ETRI).

2019 | **A Study on the Understanding of Human Situation Based on Deep Learning**
Undergraduate Research Project – Gachon University
- Proposed a multi-task learning framework combining human pose estimation and action recognition.
- Addressed real-world situational understanding using contextual cues.
- Funded by Electronics and Telecommunications Research Institute (ETRI).

Education

Mar. 2021 - Feb. 2023	**Gachon University** *M.S. in Software Engineering*
Mar. 2016 - Feb. 2021	**Gachon University** *B.S. in Department of Computer Engineering*